

```

package com.votreapp.repository;

import com.votreapp.entity.Contrat;
import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.data.jpa.repository.Query;
import org.springframework.data.repository.query.Param;

import java.util.Optional;

/**
 * Filière Bancassurance – miroir de proc_tabcrm_bq.
 * Catégories: 71,72,73,74,75,76,78,82,83,85,86,87,92
 *
 * Filtres "actif" repris du PL/SQL:
 *   a.dannulat IS NULL
 *   a.dateff IS NOT NULL
 *   NVL(a.etat,'X') <> 'I'
 */
public interface BancassuranceContratRepository extends JpaRepository<Contrat, Integer> {

    @Query("""
        SELECT c FROM Contrat c
        JOIN FETCH c.client cl
        JOIN FETCH c.category cat
        LEFT JOIN FETCH c.agence ag
        WHERE c.categorie = :numCat
              AND c.numContrat = :nadh
              AND cl.cin = :cin
              AND c.dateAnnulation IS NULL
              AND c.dateEffet IS NOT NULL
              AND (c.etat IS NULL OR c.etat <> 'I')
        """)
    Optional<Contrat> findContratActif(
        @Param("cin") String cin,
        @Param("numCat") Integer numCat,
        @Param("nadh") Integer nadh);

    static boolean isBancassurance(Integer numCat) {
        return switch (numCat) {
            case 71, 72, 73, 74, 75, 76, 78, 82, 83, 85, 86, 87, 92 -> true;
            default -> false;
        };
    }
}

```

